SHAMS UL HAQUE

Islamabad, Pakistan | shamsul9110@gmail.com

RESEARCH INTERESTS

Photocatalysis and photoelectrochemical (PEC) systems; semiconductor heterojunctions and doping; metal-oxide nanomaterials (WO_3, ZnO); green synthesis; environmental remediation; materials for energy applications.

EDUCATION

MS Chemistry, COMSATS University Islamabad *2025 – Present*
CGPA 3.7 (final semester). Thesis: Co-doped WO_3@ZnO heterojunction photocatalysts for dye degradation: synthesis, structural/optical characterization and kinetic evaluation.

BS Chemistry, Quaid-i-Azam University, Islamabad *2019 – 2024*
CGPA 3.2

PUBLICATIONS

1. **Haque, S. U.***, Nawaz, A., Farooq, E., Ullah, A., Ilyas, A., Afridi, H. I., & Ghaffar, A. (2026). Extract-mediated synthesis of ZnO nanostructures for enhanced antibacterial activity against multidrug-resistant bacteria. *Journal of Pharma and Biomedics*, 4(4), 65–81. DOI: 10.56810/jpbm.004.04.0247. (*First and corresponding author)
2. Zulfiqar, T., Irshad, I., Shah, T., Kamran, M., Shahzad, M. S., **Haque, S. U.**, & Ullah, A. (2026). Biomass-derived hierarchical porous carbon decorated with ZnO nanoparticles for enhanced electrochemical energy storage. *International Journal of Advanced Research (IJAR)*, 3(5).

RESEARCH EXPERIENCE

MS Research Student, Materials Chemistry / Photocatalysis, COMSATS University Islamabad *2025 – Present*

- Synthesize WO_3, ZnO, Co-doped WO_3 and WO_3@ZnO composites by hydrothermal and wet-chemical routes with ultrasonication-assisted mixing and calcination; vary dopant level and composite ratio to tune heterojunction behaviour.
- Interpret XRD (phase identification, Debye–Scherrer crystallite size, lattice parameters, strain), SEM–EDS (morphology, composition) and UV–Vis/Tauc band-gap data.
- Run UV–Vis photocatalytic dye-degradation studies (Beer–Lambert calibration, kinetic fitting in Origin).
- Developed a standard protocol for photocatalytic dye-degradation experiments and Excel calculators for XRD analysis, sol-gel preparation, doping and kinetics.
- Prepared a research proposal on WO_3/ZnO/PANI nanocomposites for dual-function room-temperature humidity sensing and visible-light degradation of ciprofloxacin.

Research Project: Green Synthesis of ZnO Nanostructures, Department of Chemistry, COMSATS University *2026 (published)*

- Synthesized neem (*Azadirachta indica*)-mediated ZnO (zinc acetate, NaOH, 400 °C calcination): hexagonal wurtzite, 20–25 nm, band gap ≈ 3.25 eV (UV–Vis, FTIR, XRD, SEM).
- Evaluated activity against multidrug-resistant isolates: MIC 25–100 µg/mL, MBC 50–200 µg/mL, ≥3 $\log_{10}$ reduction at 2× MIC, biofilm inhibition and disruption; ROS, membrane permeability and leakage assays indicated a multifactorial mechanism.

PROFESSIONAL EXPERIENCE

Quality Control Analyst, Vision Pharmaceuticals, Islamabad *Sep 2023 – Aug 2025*

- HPLC and FTIR analysis of drug products including IV preparations; dissolution and stability studies.
- Pharmaceutical-water testing (TOC, liquid particle count); supported dosage/weight-optimization work.
- Maintained analytical records in line with GMP and ISO guidelines.

TEACHING EXPERIENCE

Science Teacher, Cambridge Curriculum (Grade 7), Pak Turk Maarif International School, Islamabad *2025 – Present*

- Plan and deliver lessons, worksheets and slide-based activities; supervise laboratory practicals and laboratory safety.

TECHNICAL SKILLS

Synthesis: hydrothermal, wet-chemical, sol-gel, green (plant-extract) synthesis, precipitation, calcination, metal-ion doping, heterojunction composites

Characterization: XRD, SEM–EDS, UV–Vis/Tauc, FTIR, HPLC

Photocatalysis & analysis: dye-degradation studies, Beer–Lambert calibration, kinetic modelling, Origin, Excel

Biological assays: agar-well diffusion, MIC/MBC, time-kill, crystal-violet biofilm, DCFH-DA, propidium iodide

Pharmaceutical QC: TOC/LPC, dissolution, stability, GMP/ISO documentation

CERTIFICATIONS, WORKSHOPS AND SEMINARS

- Good Manufacturing Practice (GMP) Certification.
- Organic Solar Cells, Technical University of Denmark (DTU), Coursera.
- Workshop on Industry–Academia Collaboration, COMSATS University Islamabad.

REFERENCES

Available upon request.